Farha Arshi, Ph.D.

Computational Chemist | DFT and Electronic Structure Modeling | AI-Assisted Materials Discovery

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RESEARCH PROFILE

Computational and synthetic inorganic chemist working at the interface of electronic-structure modeling and experimental materials chemistry. Doctoral research combined density functional theory (DFT), frontier molecular orbital and molecular electrostatic potential analysis, and molecular docking with full hands-on synthesis and characterization of ruthenium coordination compounds and Schiff-base ligand systems. Author of five peer-reviewed publications, one book chapter and one preprint (26 citations, h-index 4), with additional experience supporting a frontier artificial-intelligence research programme as a chemistry domain reviewer. Seeking a postdoctoral appointment applying computational modeling, structure-property analysis and AI-assisted research workflows to the discovery, design and optimization of advanced functional materials. My future research focuses on integrating Density Functional Theory (DFT), machine learning, and AI-assisted computational modelling to accelerate the discovery and optimisation of advanced functional materials.

RESEARCH INTERESTS

Computational Materials Discovery | Density Functional Theory (DFT) | Electronic Structure Calculations | Computational Materials Science | AI-Assisted Molecular and Materials Design | Data-Driven Materials Design | Structure-Property Relationships | Coordination and Organometallic Chemistry | Ruthenium Coordination Chemistry | Advanced Functional Materials | Materials Synthesis and Characterization | Quantum Chemistry | Virtual Screening and Computational Workflow Design

EDUCATION

Ph.D., Chemistry 2020 - 2026
University of Lucknow, Lucknow, India (NAAC A++ accredited)

Thesis: Synthesis, Characterization and Biological Evaluation of Ruthenium Complexes with Different Amines. Supervisor: Dr. Ashok Kumar Singh.

M.Sc., Chemistry 2016 - 2018
University of Lucknow, Lucknow, India

B.Sc., Chemistry, Zoology and Botany 2013 - 2016
University of Lucknow, Lucknow, India

RESEARCH EXPERIENCE

Doctoral Researcher - Computational and Inorganic Materials Chemistry 2020 - 2026
University of Lucknow, Lucknow, India

- Designed and synthesized Schiff-base ligands and a structurally diverse library of ten new ruthenium(III) and half-sandwich ruthenium(II)-arene complexes through multistep inorganic and organometallic synthesis.
- Performed density functional theory calculations in Gaussian and GaussView, including geometry optimization, frontier HOMO–LUMO analysis, energy-gap and global reactivity descriptors, and molecular electrostatic potential (MEP) mapping, alongside experimental synthesis and characterization to explain electronic structure, reactivity, and structure–property relationships.
- Conducted molecular docking and binding-interaction studies using AutoDock and BIOVIA Discovery Studio, together with ADME and drug-likeness, to evaluate binding affinity, analyse protein–ligand interactions, and complement experimental biological studies.
- Characterized all compounds using FT-IR, UV-Vis, NMR (1H and 13C), mass spectrometry, single-crystal X-ray diffraction and CHNS elemental analysis, and reconciled experimental data against the computed models..
- Published research in 5 peer-reviewed international journals, contributing to 26 citations (h-index: 4), with 1 book chapter, 1 preprint, and 1 manuscript currently under revision.
- Coordinated biological evaluation with partner laboratories, authored five peer-reviewed papers and a book chapter, presented at national and international conferences, and mentored junior researchers in synthesis, characterization and computational workflow setup.

Trainee - Computational Approaches to Drug Design and Development (CADD) Jan 2019 - Mar 2019
CSIR - Central Drug Research Institute (CDRI), Lucknow, India

- Completed an intensive eight-week certificate programme covering molecular modeling, virtual screening, molecular docking and bioinformatics tools applied to computational discovery workflows.

SCIENTIFIC COMPUTING AND AI PROJECT EXPERIENCE

Chemistry Domain Reviewer - Scientific AI Evaluation Project (Project Planck) 2026 - Present
Handshake AI (remote)

- Developed graduate- and doctoral-level chemistry reasoning tasks for the evaluation of scientific AI system.

- Validate structures computationally using RDKit with canonical SMILES and InChI representations, and support every item with open-access, DOI-referenced literature.
- Validated scientific content using peer-reviewed literature and contributed to quality assurance of chemistry-focused AI evaluation tasks.

TECHNICAL SKILLS

Computational Chemistry: Density functional theory (DFT); geometry optimization; frontier molecular orbital (HOMO-LUMO) analysis; molecular electrostatic potential (MEP) mapping; global reactivity descriptors; Gaussian; GaussView. Rdkit, Python

Molecular Modeling and Docking: AutoDock; BIOVIA Discovery Studio; structure-based design; binding-energy and interaction analysis; virtual screening; ADME and drug-likeness prediction; QSAR.

Spectroscopic and Structural Characterization: FT-IR; UV-Vis spectroscopy; NMR (1H, 13C); mass spectrometry; CHNS elemental analysis; interpretation of spectral and crystallographic data.

Experimental and Synthetic Techniques: Schiff-base ligand synthesis; ruthenium and transition-metal complex synthesis; multistep inorganic and organometallic synthesis; reflux and inert handling; column chromatography (silica and alumina); crystallization, recrystallization and purification.

Data Analysis and Software: OriginPro; ChemDraw; Microsoft Excel; scientific data analysis and figure preparation.

Research and Professional: Scientific writing and peer-reviewed publication; research proposal development; conference presentation; mentoring and supervision of junior researchers; interdisciplinary and multi-laboratory collaboration.

RESEARCH OUTPUT

5 peer-reviewed journal publications | 1 book chapter | 1 preprint | 1 manuscript under revision | 26 citations | h-index 4 | 3 conference presentations (1 international).

PEER-REVIEWED PUBLICATIONS

1. Exploring Schiff Base and its Palladium Complexes: Synthesis, Characterization, Antimycobacterial Activity, DFT, Molecular Docking and ADME Studies. *Journal of Molecular Structure (2026).* DOI: 10.1016/j.molstruc.2025.143704

2. New Ru(III) 2,6-Bis(2-Benzimidazolyl)Pyridine Complexes Bearing p-Sub-Benzyl Thiosemicarbazones Schiff Base: Synthesis, Characterization, DNA Binding and Anti-cancer Activity. *Chemistry - An Asian Journal (2025).* DOI: 10.1002/asia.202500059

3. Design, Synthesis, and Biological Insights of Half-Sandwich Ruthenium-Arene Schiff Base Complexes: Molecular Docking and DFT. *ChemistrySelect (2025).* DOI: 10.1002/slct.202404895

4. Synthesis, DFT Calculation, DNA Binding and Biological Evaluation of Mononuclear Ru(III) Complexes with 2,6-Bis(2-benzimidazolyl)pyridine Bearing p-Substituted Heterochalcones. *Russian Journal of General Chemistry (2023).* DOI: 10.1134/S1070363223020202

5. Design, Synthesis, Theoretical, Spectroscopic and Molecular Docking Studies of Ruthenium and Zinc Complexes and their Antimycobacterial Study. *Asian Journal of Chemistry (2022).* DOI: 10.14233/ajchem.2022.23867

Preprint and Manuscript in Progress

. Ampicillin-Derived Ruthenium Schiff Base Complexes as Emerging Anticancer and Antimicrobial Candidates: Synthesis, Characterization, DFT, ADME and Molecular Docking. *Preprint, SSRN (2024).* DOI: 10.2139/ssrn.5512074

- Fe(III)-bipyridine thiosemicarbazone Schiff-base complexes - synthesis, characterization, DFT and molecular docking. Manuscript under revision, ChemistrySelect.

BOOK CHAPTER

- Arshi, F., & Singh, A. K. (2022). Synthesis, Characterization, Molecular Docking and Biological Evaluation of Ruthenium Complexes. In *Advancement in Chemical Sciences* (Chapter 7, pp. 97-118). Aryabhat Publication House. ISBN: 978-93-95463-01-0.

CONFERENCE PRESENTATIONS AND WORKSHOPS

- **Poster Presentation - 30th ISCB International Conference (2025), International Society for Chemical and Biological Sciences.** *Comprehensive Exploration of Ampicillin-Schiff Base Metal Complexes: Synthesis, DFT Insights, Molecular Docking, and Biotherapeutic Applications.*
- **Oral Presentation - International Conference on Multidiscipline Academic Research and Innovation, ICMARI (2022).** *Synthesis, Characterization, Molecular Docking and Biological Evaluation of Ruthenium Complexes.*
- **Oral Presentation - National Seminar on Sustainable Development and Environmental Protection (2023), Netaji Subhash Chandra Bose Government Girls P.G. College, Lucknow.** *Synthesis, Spectral Studies and Molecular Docking of Thiosemicarbazide Ligands and Ruthenium Complexes.*
- **Computational Approaches to Drug Design and Development (CADD)** - eight-week certificate programme, CSIR - Central Drug Research Institute (CDRI), Lucknow, 2019.
- Good Laboratory Practice (GLP) - workshop.
- **HPTLC: Technique and Applications - virtual workshop**, Anchrom Enterprises (I) Pvt. Ltd. and Smriti College of Pharmaceutical Education.